Leeds Trinity University | School of Computing and Creative Industries

Title - Granular Data, Hidden Risks: The Ethics of Smart Meter IoT Data Collection by UK Energy Firms

Assessment 2: Investigative Report

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1. Introduction

The United Kingdom is currently in the process of implementing one of the most extensive smart meter rollouts across the world. By the end of 2025, the government had already set a target for energy suppliers to roll out smart meters to all household and small business consumers, with more than 35 million already in progress (BEIS, 2022). Smart meters are Internet of Things (IoT) devices that enable automatic communication and transmission of electricity and gas usage information to energy suppliers via the Data Communications Company (DCC), a safe and secure network. Instead of old dials that need reading by hand, they update consumption levels automatically every half hour using digital signals. That constant flow builds an exact picture showing how people use utilities throughout each day without guesswork or delays.

Some people say smart meters bring clear wins - bills match actual use, less power gets wasted, pricing can shift with demand, and they help Britain hit its climate goals. However, there are many ethical issues that have yet to be fully explored. This report is intended to discuss the way energy companies use personal data through smart meter technology and its effects on personal data privacy, legal rights, and economic justice. Smart meter technology is useful for many reasons, but its ability to gather detailed personal data and its potential for further exploitation by energy companies and government agencies are problematic.

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2. How Smart Meters Collect and Transmit Data

Smart meters are made of two meters, one for electricity and one for gas, and they come with an In-Home Display (IHD), which shows you how much energy you are using at any given moment. Your electricity use is shown on the IHD every ten seconds, but gas information updates every thirty minutes (Smart Energy GB, 2024). These meters send their readings to your energy supplier through their own wireless connection, transmitting readings either daily or at half-hourly intervals, depending on consumer settings and supplier requirements.

Each smart meter has a unique Meter Point Administration Number (MPAN). And when you connect that number to how much energy you have used, it is considered your personal data according to the law in the UK (ICO, 2025). The Information Commissioner's Office (ICO) says that using your MPAN with half-hourly details of your energy use is personal data, and so is fully covered by the UK General Data Protection Regulation (UK GDPR) and the Data Protection Act 2018 (ICO, 2025).

The information coming from smart meters does not just go from you to your supplier. Distribution Network Operators (DNOs) can get hold of half-hourly figures, but they are combined, and names are removed, and they must follow privacy plans approved by Ofgem. This is to help them manage the power network more effectively. Also, other companies with permission - energy services or sites where you compare and switch suppliers, for example - can get your individual energy use details, but only if you specifically say they can. This creates a complex ecosystem of data actors, each with differing levels of access, regulatory oversight, and commercial incentives.

The smart meter ecosystem therefore involves four stakeholder groups with conflicting interests: consumers who face privacy risks they may not fully understand; energy firms who stand to gain commercially from granular data analytics; regulators such as Ofgem and the ICO whose mandates — market efficiency versus data protection — directly conflict; and computing professionals who carry independent ethical obligations under the BCS and ACM codes regardless of employer priorities.

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3. Ethical Analysis

3.1 Privacy and the Inference Problem

What worries people most about smart meters is not just tracking power usage - it is how much personal detail gets exposed. Studies show readings taken every thirty minutes can suggest daily routines: waking times, sleeping habits, presence at home, appliance choices, sometimes even clues about physical condition or way of living (National Grid, n.d.). According to the Open Rights Group (2023), fine-grained electricity information might reveal if someone stayed in overnight, rested properly, or followed expected conduct tied to welfare rules - insights potentially accessed by lenders, coverage providers, rental owners, or public authorities.

This idea gains clarity by applying Helen Nissenbaum’s concept of contextual integrity - ethical data sharing depends on whether information moves in ways that fit the setting where it first emerged (Nissenbaum, 2004). While people accept meters being placed in homes to calculate utility costs, such permission does not extend to their detailed behavioural traces fuelling targeted advertising, eligibility checks, or financial evaluations. Once consumption records shift beyond invoicing into unrelated areas, expectations break down. Such crossings disrupt alignment with social norms, weakening faith in how personal details are managed.

3.2 Consent, Autonomy, and the Opt-In Illusion

Half-hourly smart meter data in the UK falls under the Data Access and Privacy Framework (DAPF), where energy providers must secure explicit permission before using it beyond standard billing tasks - so states Ofgem/BEIS (2018). Though designed to uphold individual choice, real-world application reveals shortcomings. According to Energy UK (2025), responding during a national review of smart data plans, true understanding rarely happens

because explaining rules clearly feels overwhelming - even experts admit clarity may be out of reach. In practice, the consumer tends to agree without necessarily understanding the implications, while the energy suppliers differ in the level of informed consent from the consumers. The level of control that is evident on paper is rarely seen in practice.

From a Kantian viewpoint, the exploitation of personal information without true informed consent treats consumers not as rational agents to be respected, but as a tool for achieving commercial success (Kant, 1785/1998). The substantial information gap between large energy organisations and individual consumers makes this concern more serious, given that large organisations have analytical capacity that most individuals cannot effectively scrutinise or dispute.

3.3 Consequentialist Analysis — Weighing Harms Against Benefits

A consequentialist framework, most associated with Mill and Bentham, judges actions by their outcomes — specifically, whether they produce the greatest good for the greatest number (Mill, 1863/2001). Applied to smart meter data collection, this produces a genuinely contested picture.

The aggregate benefits are real: more efficient grid management, renewable energy integration, and time-of-use tariffs that could reduce household costs. If granular data collection meaningfully accelerates decarbonisation, a utilitarian calculus might appear to justify the privacy costs involved.

However, two problems undermine this conclusion. First, harms and benefits are distributed unequally — risks fall disproportionately on vulnerable and digitally excluded households, while financial gains accrue primarily to energy firms and technology companies. A utilitarian analysis that ignores this distributional dimension is misleading. Second, the long-term consequences of normalising behavioural surveillance infrastructure are difficult to predict. As the Open Rights Group (2023) demonstrated, once a half-hourly data collection infrastructure exists, its repurposing for credit assessment or welfare fraud detection becomes technically trivial. A responsible consequentialist approach must therefore ask whether the aggregate benefits could be achieved through less privacy-invasive means before justifying the current scale of collection.

3.4 The ICO–Ofgem Conflict and Regulatory Fragmentation

One of the more disturbing aspects of this problem is the direct conflict that exists between the two most relevant regulatory bodies in the UK. In December 2025, for example, the ICO lodged a formal objection to Ofgem's proposed requirement for half-hourly settlement for all consumers, based on the fact that this would directly contradict the opt-in requirement outlined in the DAPF and effectively deny consumers a choice regarding the collection of personal data (Pinsent Masons, 2025). The ICO warned that, in this way, suppliers could profile individuals and make decisions based on this information, a practice that they themselves had previously warned against and attempted to mitigate through the DAPF framework.

As such, this regulatory conflict points to a greater accountability gap. While Ofgem's role is to promote efficient energy markets and support decarbonisation, the ICO's role is to protect personal data rights. However, when these interests conflict, no superior authority exists to protect consumers in vulnerable situations. According to Energy UK in 2025, handing data beyond the sector can shift oversight away from Ofgem entirely - yet exactly how involved the ICO becomes afterwards remains uncertain. Without clear direction, people face real consequences, especially if they are already struggling. From a stakeholder perspective, this

regulatory fragmentation leaves consumers — particularly vulnerable ones — caught between two powerful institutions whose mandates do not align, with no clear mechanism for redress.

3.5 Fairness, Economic Harm, and Vulnerable Households

Smart meter data collection raises important concerns about fairness and economic inequality. The government consultation for its energy smart data scheme (DESNZ, 2025), while noting the potential for smart data sharing to bring benefits such as improved tariff comparison and retrofitting advice, also recognises that barriers to participation will disproportionately impact vulnerable consumers. Digital exclusion was cited as a barrier by 82% of those responding to the consultation, and a lack of understanding was cited by 53%. Elderly consumers, those on low incomes, and those with low levels of digital literacy are those who are most vulnerable to the potential harms of smart data sharing.

The Open Rights Group (2023) revealed that the UK government had previously proposed retaining smart meter data for ten years and sharing it with credit reference agencies, local authorities, and debt recovery agencies for fraud detection purposes — a plan reversed only after significant public opposition. That plan made clear how personal usage details might be used against people facing money troubles. Energy suppliers currently retain the capacity to develop differential pricing models based on historical consumption profiles, raising concerns about algorithmic discrimination. Where pricing systems penalise consumption patterns that vulnerable households cannot easily alter — such as daytime electricity use by elderly or disabled residents — the outcome is structurally unjust. This represents a form of data-driven inequality that existing regulatory frameworks do not adequately address.

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4. Legal and Regulatory Framework

In the UK, the data collected through smart meters is subject to a complex web of regulations. At the base is the UK GDPR and Data Protection Act 2018, which state that personal data should be processed lawfully, fairly, and transparently for specific purposes and in a manner that ensures appropriate security. Next is a series of sectoral regulations, including Ofgem's Data Access and Privacy Framework, the Smart Energy Code, and the Data (Use and Access) Act 2025.

The Data (Use and Access) Act 2025 is a notable legislative step. It grants legislative powers to establish Smart Data Schemes. Such schemes have the power to force data holders, including energy providers, to share consumer data with third parties and regulators (Kennedys Law, 2025). Even though the intention behind the schemes is to ensure consumer benefit and competition in the market, it also increases the limits of mandatory consumer data sharing beyond what the consumer might think when they agree to install a smart meter. The Act also states that data holders can be obliged to comply with standards and codes of conduct and interoperability. However, the legislative powers of the ministers in the Act give the Secretary of State significant powers to determine the sharing of data and its conditions.

The professional obligations of computing practitioners in this space also warrant attention. The BCS Code of Conduct requires that BCS members have due regard for public health, privacy, security, and well-being. It also requires them to promote equal access to the benefits of IT. This means that for engineers and data scientists designing smart meter data systems, it requires them to undertake Data Protection Impact Assessments before launching new data collection capabilities and to be aware of organisational pressure to extend the use of data beyond its originally consented purpose. The ACM Code of Ethics also requires computing

professionals to 'avoid harm' and 'protect the privacy of others.' This requirement is relevant to the work of designing a system that has the capability to make inferences about intimate personal patterns from energy consumption data (ACM, 2018). These codes require that computing professionals understand that their ethical obligations in this space are not just about compliance but also a personal requirement for every computing practitioner involved in smart meter data systems.

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5. Wider Societal, Economic, and Environmental Implications

The societal implications of large-scale smart meter data collection extend well beyond the individual household. At an aggregated level, smart meter data has substantial public interest value in enabling better management of the grid, facilitating renewable energy integration, and enabling detailed research into energy poverty. The government's smart data consultation (DESNZ, 2025) identified many legitimate use cases ranging from better tariff comparison to targeted advice on retrofits.

However, these advantages must be set against the potential costs of what has been termed the 'surveillance economy,' whereby value is extracted from personal data by powerful organisations (Zuboff, 2019). The energy sector's drive towards smart data systems mirrors other sectors, such as open banking and telecommunications, to build out a growing infrastructure for the exploitation of personal household data. With the growing capabilities of artificial intelligence and machine learning technologies, the potential for re-identification of individuals from anonymous data sets (Open Energy, n.d.) also grows, as studies have shown that aggregated energy data can be de-anonymised under certain circumstances.

From the point of view of the environment, smart metering is seen to enable the decarbonisation of energy supplies. This is true and significant. However, the case for smart meters in terms of the environment does not extend to the loss of privacy. When assessing the impact of modern technologies, it is important that social and ethical costs are considered. This is the essence of responsible innovation. Concerning smart meters in particular, this entails making sure that the infrastructure we develop to enable decarbonisation does not simultaneously enable the emergence of a surveillance economy. A truly sustainable energy transition must also be socially just as well as environmentally effective.

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6. Recommendations

From the analysis above, this report makes the following recommendations to policymakers, regulators, and computing professionals:

- **Strengthen consent mechanisms:** The government should direct energy suppliers to explain data sharing in plain English at the time of smart meter installation and at every subsequent change in data use. The consent should be freely given and withdrawable. The ICO should have the power to enforce this requirement on both the private and public sectors. This directly addresses the Kantian concern that consumers are currently treated as means rather than ends.
- **Resolve the ICO-Ofgem conflict:** The government should resolve the conflict of regulatory powers in the event of data protection and energy sector goals clashing. The privacy rights of consumers should not be subordinated to energy efficiency and commercial goals without parliamentary scrutiny. From a consequentialist perspective,

regulatory clarity is essential to ensure that aggregate benefits do not come at disproportionate cost to the most vulnerable.

- **Protect vulnerable consumers:** Any smart data initiative should have specific provisions for vulnerable consumers, including the digitally excluded and the financially vulnerable. This should include the prohibition of using consumption data for credit checks and benefit fraud without primary legislation. Computing professionals have a particular responsibility here — their design choices directly determine whether vulnerable stakeholders are protected or exposed.
- **Ensure privacy by design:** Computing professionals designing smart meter data systems should be made to conduct Data Protection Impact Assessments before introducing new data collection or sharing. They should also ensure data minimisation, i.e., collecting only the data necessary for the proposed use.
- **Strengthen third-party accountability:** The third parties using smart meter data through the smart data scheme should be made accountable through robust audit and enforcement procedures. There should also be redress procedures for consumers. The ICO should have specific oversight powers over the use of data by third parties, not only the suppliers.

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7. Conclusion

Smart meters are, in fact, a technology of genuine importance to the UK's energy future. The ethical basis of smart meter data collection, however, is grounded in a situation that is currently unstable, with consent processes that are hard to understand, regulatory frameworks that are in direct conflict with one another, and escalating business and government interests in using the data gathered in the home for purposes never envisaged by the consumer.

The ICO's December 2025 warning on the mandatory imposition of half-hourly settlement, the Open Rights Group's exposure of government surveillance plans, and the complex tensions inherent in the Data (Use and Access) Act 2025 all point to the same conclusion: the pace of smart meter technology has far outstripped the development of the ethical and regulatory infrastructure to protect the consumer.

To bridge this gap, not only is better regulatory enforcement needed, but also a cultural transformation in the energy sector and computing science alike, one in which concepts such as privacy and consent are not obstacles to be overcome, but essential enabling environments for innovation. For computing science professionals, this is already spelt out in the BCS Code of Conduct, which emphasises that public welfare and privacy protection are not only not optional, but essential to being a professional.

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Appendices

This assignment used generative AI in the following ways to complete the Assignment: brainstorming, research, planning, and editing.





